REACTION-WHEEL INVERTED PENDULUM

LQG stabilization on embedded hardware • Mechanical design • State estimation • Real-time control

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RP2350 • C • MATLAB • Python

LQR • Kalman • Embedded Controls

Project Overview

Designed and built a reaction-wheel inverted pendulum from first principles, including the mechanical assembly, motor drive electronics, embedded firmware, state-space model, discrete LQR controller, and Kalman observer. The final system catches near upright and maintains closed-loop balance while estimating unmeasured states in real time.



Performance

Over 9 experimental trials

9/9

Successful catches

0.227°

Mean steady-state angle RMSE

0.072°

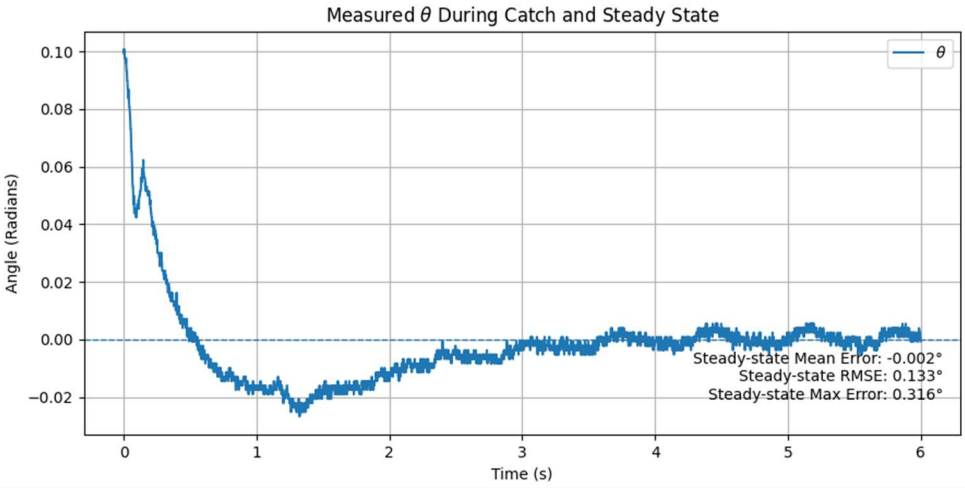
Mean angle-estimation RMSE compared with ground truth encoder measurements

43.8%

Mean reduction in flywheel-speed variation

(Steady-state is measured from t = 3s onward)

Closed-Loop Performance



- Catch transient demonstrates recovery into the upright operating region.
- Steady-state metrics quantify actual balance performance, not only estimator agreement.
- Controller was tuned on real hardware after diagnosing excessive actuator-effort penalty and model/actuator nonlinearities.

CONTROL, ESTIMATION & IMPLEMENTATION

System Architecture

Sensors	Estimator	Controller	Plant
Pendulum hall encoder Flywheel quadrature encoder Current sensing amplifier	Discrete Kalman filter State estimate $\hat{x}$	Discrete LQR $u = -K\hat{x}$	Geared DC motor Reaction wheel Pendulum body

Control Model

State Vector

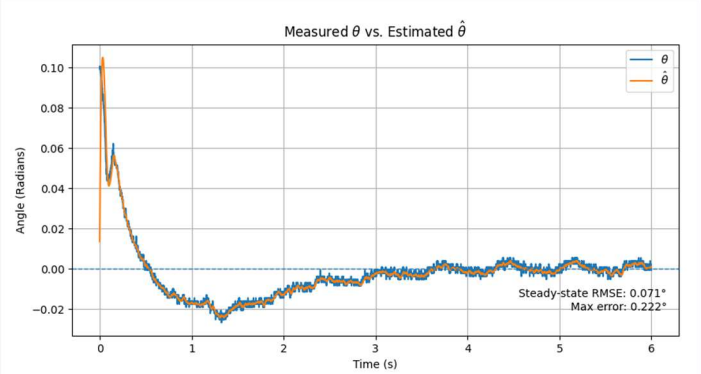
$$x = [\,\theta,\ \dot{\theta},\ \rho,\ i\,]^T$$

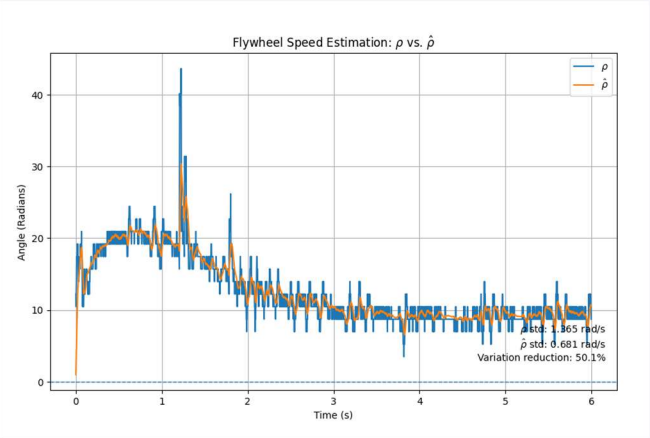
θ: pendulum angle • ρ: flywheel speed • i: motor current

Implementation

- 2 ms discrete control timestep
- RP2350 / Raspberry Pi Pico firmware in C
- MATLAB controller + estimator design
- Python USB logging and performance analysis

Estimator Performance





Engineering Iteration

Observed behavior	Diagnosis / analysis	Design change
Large limit cycle near upright	Controller too conservative; real actuator friction/deadzone	Reduced actuator-effort penalty and retuned LQR
Encoder zero did not match physical upright	Measured equilibrium bias with repeated static samples	Added calibrated mechanical angle offset
Noisy / quantized flywheel-speed measurement	Compared raw ρ with model-based estimate	Retuned Kalman covariance for smoother state feedback

[Click here to see a short video of the pendulum!]